Model Disagreement-Based Reliable Crop Recommendation Using Machine Learning

1st Supti Das Medha
(Student ID: 08123105101041)
Dept. of CSE, BAUET
Natore, Bangladesh
suptidasmedha@gmail.com

***Abstract*—Crop recommendation systems usually report a single accuracy figure, which says little about how far an individual recommendation can be trusted. This paper studies the agreement between four heterogeneous classifiers, a Decision Tree and Support Vector Machines (SVMs) with linear, polynomial and RBF kernels, as a reliability signal, and asks whether disagreement cases have higher prediction errors than agreement cases. On a public dataset of 2,200 samples covering 22 crops and seven soil and climate features, the models reach 98.64%, 97.73%, 96.82% and 92.05% hold-out accuracy (Decision Tree, linear, RBF and polynomial SVM; 440 test samples). All four models agreed on 401 test samples (91.14%); only 2 of these predictions were wrong (0.50% error), whereas at least one model erred on every one of the 39 disagreement samples; the error rate of every individual model is also higher on disagreement samples (e.g., 10.26% versus 0.50% for the Decision Tree). Accepting the majority label only when at least three of four models agree gives 97.95% coverage at 98.61% accuracy (1.39% risk), and requiring unanimity gives 91.14% coverage at 0.50% risk. The majority vote does not exceed the best single model, so we present agreement as a low-cost abstention and review-flagging mechanism rather than an accuracy booster. We also discuss the limits of a single split on a small, clean benchmark.**

Keywords—crop recommendation, model disagreement, classifier agreement, majority voting, selective prediction, coverage–risk, support vector machine, decision tree.

I. INTRODUCTION

Selecting the appropriate crop for a specific area relies on soil nutrients, pH, temperature, moisture, and precipitation, and an incorrect selection directly impacts production and revenue. Consequently, machine learning has found extensive use in crop recommendation, usually through training classifiers based on soil and climate data and assessing performance on a reserved dataset .

A lone accuracy metric, however, does not inform a farmer or an advisor about which specific recommendations to rely on. When multiple independently trained models exist, their agreement or disagreement on a specific input provides additional information with minimal extra cost: if models with different inductive biases all produce the same result, the recommendation is likely more trustworthy than when they diverge. This concept pertains to ensemble learning and classification with a reject option, yet it is seldom quantified in the context of crop recommendation.

This paper thus focuses on: Are prediction errors greater in disagreement cases than in agreement cases? In order to address this question, this paper offers the subsequent contributions:

1. Evaluates four classifiers (Decision Tree and linear, polynomial, and RBF SVMs) on a 22-class crop recommendation dataset using one method: 5-fold cross-validation on the training data .
2. Divides the test set based on model consensus and assess the distribution of prediction errors across the agreement and disagreement groups.
3. Assesses agreement-gated majority voting with clear coverage and risk and fairly compare it to the individual models.
4. Outlines what these findings endorse and do not endorse, including the statistical relationship between the agreement indicator and the count of model errors.

II. RELATED WORK

Crop suggestion. Lokhande and Dixit created a recommendation system utilizing a dataset of 2,200 samples related to soil and climate through decision trees, SVMs, and random forests [1]. Singh et al. evaluated K-nearest neighbors, decision trees, and random forests, revealing a 99.59% accuracy using a random forest classifier [2]. Afzal et al. suggested a combination of random forest and extreme gradient boosting for 22 crops, employing hyperparameter optimization and K-fold cross-validation, achieving 98% accuracy [3]. These studies concentrate on identifying the model that achieves the greatest accuracy; they do not examine when specific predictions can be relied upon.

Conjunctions and variety. Voting to combine classifiers is a common method to enhance or stabilize predictions [4], with the advantage relying on the diversity of the ensemble members [6]. The focus here is not just on the vote itself but on the underlying agreement pattern that supports it.

Discriminative categorization. Chow defined the balance between error and rejection [5]. Selective classification examines the balance between coverage and risk for classifiers that can opt out [7], and this has been developed for deep networks as well [8]. We utilize the same coverage–risk perspective, but derive the abstention signal from differences among basic classical models rather than from a confidence score of a single model.

III. DATA AND METHODOLOGY

A. Data Collection

Utilize the public Crop Recommendation dataset [9]: it contains 2,200 samples, seven numerical features, and 22 crop labels, each with precisely 100 samples per category. Table I presents the characteristics along with their observed ranges. There were no duplicate rows or missing values detected.

TABLE I
DATASET FEATURES AND OBSERVED RANGES

Feature	Details	Minimum	Maximum
N	Nitrogen content in soil	0	140
P	Phosphorus content in soil	5	145
K	Potassium content in soil	5	205
Temperature	Temperature(°C)	8.83	43.68
Humidity	Relative humidity(%)	14.26	99.98
ph	Soil pH	3.50	9.94
Rainfall	Rainfall(mm)	20.21	298.56

B. Division of Data and Preprocessing

The dataset was divided once into 1,760 training samples and 440 test samples (80/20, random seed 42). The division was arbitrary and not stratified, resulting in class support in the test set ranging from 11 to 27 samples. For the SVMs, the features were normalized to have zero mean and unit variance utilizing a scaler within a pipeline, ensuring that the scaler is applied solely on the training data (and on the training folds exclusively during cross-validation). The Decision Tree was trained using the raw features, as decision trees are unaffected by feature scaling.

C. Classifiers

Utilized four classifiers from scikit-learn with their default configurations (Table II); hyperparameter tuning was not performed. We selected SVMs and a Decision Tree due to their distinct methods of data partitioning—SVMs employ mathematical boundaries, whereas decision trees apply rule-based partitions—making their comparison significant.

TABLE II
CLASSIFIER CONFIGURATIONS

Model	Configuration
SVM(Linear)	Standardization + SVC, linear kernel, C = 1
SVM(Linear)	Standardization + SVC, polynomial kernel, degree 3, C = 1
SVM(Linear)	Standardization + SVC, RBF kernel, C = 1, gamma = "scale"
Decision Tree	Fully grown tree, random seed 42

D. Evaluation Protocol

Each model was first evaluated using stratified 5-fold cross-validation on the training data, with shuffling and a random seed of 42 [13]. The mean accuracy, macro-averaged precision, recall, and F1-score were recorded. After this step, each model was trained again using the entire training set and tested on the same 440 test samples. The test data were kept separate throughout model selection and tuning to ensure a fair evaluation. All four models were tested on exactly the same samples.

E. Agreement Analysis and Majority Voting

For each test sample,compared the predicted labels from all four models. If all four models predicted the same label, the sample was considered an agreement sample. Otherwise, it was treated as a disagreement sample. We also counted how many models supported the most frequently predicted label, denoted by (k), where $(k \in \{4,3,2\})$.

The system accepted a prediction when at least three models agreed on the same label $((k \geq 3))$. However, when only two models supported the most frequent label $((k = 2))$, such as a 2–2 or 2–1–1 split, the system did not accept the prediction and instead abstained.

Let (N)represent the total number of test samples, (n_{acc}) the number of accepted predictions, and (n_{err}) the number of incorrect predictions among the accepted samples. We calculated coverage and risk using the following formulas:

$$Coverage = \frac{n_{acc}}{N}$$

$$Risk = \frac{n_{err}}{n_{acc}}$$

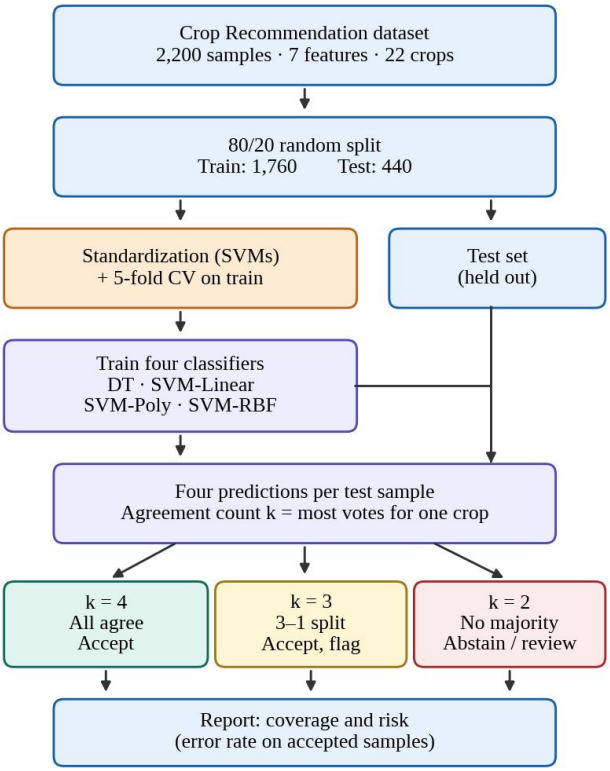


Fig. 1. Workflow of the proposed model disagreement-based crop recommendation. Four classifiers are trained on the same training data, their predictions on the held-out test set are compared, and the agreement count k decides whether a recommendation is accepted, flagged or withheld.

IV. RESULTS

A. Individual Models

Table III reports cross-validation and hold-out results. In cross-validation the RBF SVM and the Decision Tree both reach 98.47% accuracy, and the polynomial SVM is clearly lowest (93.18%). On the test set the Decision Tree is best (98.64%, 6 errors), followed by the linear SVM (97.73%, 10 errors), the RBF SVM (96.82%, 14 errors) and the polynomial SVM (92.05%, 35 errors). The RBF SVM drops 1.65 points from cross-validation to test, whereas the Decision Tree gains 0.17. Because one test sample is worth 0.23 points, the gaps among the top three models correspond to only a few samples (Section V-D).

Table III

CROSS-VALIDATION (CV, TRAINING SET) AND HOLD-OUT TEST RESULTS

Model	CV Acc.	CV F1	Test Acc.	Test P	Test R	Test F1
SVM (Linear)	0.9818	0.9819	0.9773	0.9776	0.9785	0.9771
SVM (Poly)	0.9318	0.9347	0.9205	0.9371	0.9233	0.9224
SVM (RBF)	0.9847	0.9845	0.9682	0.9677	0.9695	0.9666
Decision Tree	0.9847	0.9844	0.9864	0.9861	0.9873	0.9864

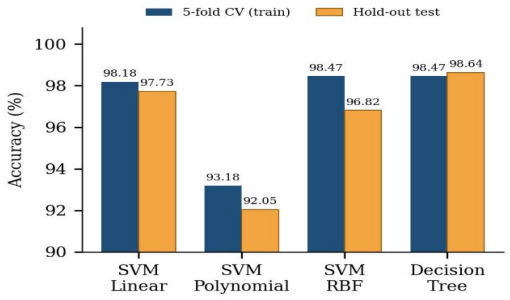


Fig. 2. Mean 5-fold CV accuracy and hold-out test accuracy per model. The vertical axis starts at 90% to make differences visible

Per-class reports show that the main confusion for the linear SVM is between rice and jute (rice recall 0.79, jute precision 0.85): five of its ten errors are rice–jute confusions. For the polynomial SVM, 16 of 35 errors are predictions of "maize", which suggests a systematic bias of the untuned model rather than random noise.

B. Agreement and Disagreement

All four models agreed on 401 of the 440 test samples (91.14%), and disagreed on 39 (8.86%). Table IV shows how errors are distributed. Within the agreement group only 2 samples were wrong (all four models agreed on a wrong label), a sample error rate of 0.50%. Within the disagreement group, at least one model was wrong on every sample by definition, with 1.46 wrong models on average. In total, 41 test samples had at least one erring model, and 39 of them (95.1%) fall in the disagreement group. No disagreement sample had all models wrong: at least one model was correct in all 39. The per-model correct counts on these 39 samples were 35 (Decision Tree), 31 (linear), 27 (RBF) and 6 (polynomial).

Table IV

ERROR DISTRIBUTION BY AGREEMENT (440 TEST SAMPLES)

Group	Samples	With ≥1 error	Mean wrong models	Error rate
Agreement	401	2	0.020	0.50%
Disagreement	39	39	1.462	100%

The "at least one model wrong" criterion is partly circular, so we also compare each model's own error rate across the two groups (Table V). Every model is wrong more often on disagreement samples: 10.26% versus 0.50% for the Decision Tree, 20.51% versus 0.50% for the linear SVM, 30.77% versus 0.50% for the RBF SVM and 84.62% versus 0.50% for the polynomial SVM. Within the agreement group each model's errors are the same 2 samples on which all four models agreed on a wrong label. This answers the research question affirmatively: disagreement cases have clearly higher prediction error.

TABLE V

PER-MODEL ERRORS BY AGREEMENT GROUP

Model	Agreement (n = 401)	Disagreement (n = 39)
SVM (Linear)	2 (0.50%)	8 (20.51%)
SVM (Poly)	2 (0.50%)	33 (84.62%)
SVM (RBF)	2 (0.50%)	12 (30.77%)
Decision Tree	2 (0.50%)	4 (10.26%)

A Welch t-test on the per-sample number of erring models gives $t = -12.41$ ($p < 10^{-14}$). We report this only descriptively. The agreement indicator and the error count are linked by definition: a unanimous sample has 0 or 4 erring models, whereas a disagreement sample has at least one. The test therefore cannot show that disagreement is an independent predictor of error; the informative quantities are the group-wise error rates in Table IV.

C. Majority Voting and Coverage–Risk

Table VI breaks the majority vote down by agreement strength. Unanimous samples ($k = 4$) are 99.50% correct under the majority label; samples with a 3–1 split ($k = 3$) drop to 86.67%, with 4 of 30 majority labels wrong. The 9 samples with $k = 2$ have no majority and are excluded. At least one model was correct on each of these 9 samples, and the Decision Tree was correct on 7 of them.

TABLE VI

MAJORITY-VOTE CORRECTNESS BY AGREEMENT STRENGTH

Agreement	Samples	Correct	Wrong	Accuracy
4 models agree	401	399	2	99.50%
3 models agree	30	26	4	86.67%
No majority (k = 2)	9	–	–	abstain

Table VII converts this into coverage and risk, and Fig. 2 plots the trade-off together with the single models, which have 100% coverage. Accepting only when all four models agree gives 91.14% coverage at 0.50% risk. Accepting when at least three agree raises coverage to 97.95% at 1.39% risk (98.61% accuracy; 425 of 431 correct).

Table VII

Policy	Accepted	Coverage	Errors	Risk
Unanimous (k = 4)	401	91.14%	2	0.50%
Majority (k ≥ 3)	431	97.95%	6	1.39%
Decision Tree alone	440	100%	6	1.36%
SVM (Linear) alone	440	100%	10	2.27%
SVM (RBF) alone	440	100%	14	3.18%
SVM (Poly) alone	440	100%	35	7.95%

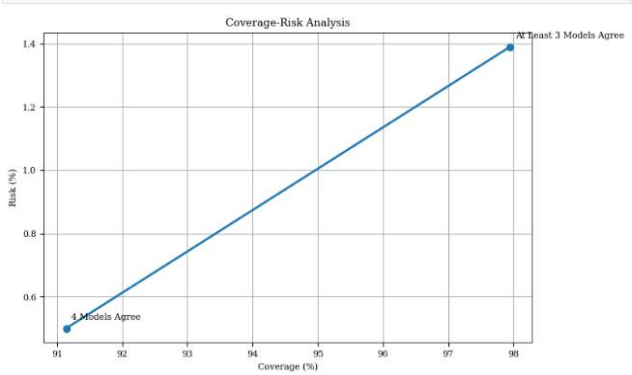


Fig. 3. Coverage–risk trade-off of agreement-gated majority voting. The two points correspond to accepting predictions only when all four models agree (91.14% coverage, 0.50% risk) and when at least three agree (97.95% coverage, 1.39% risk)

V. Discussion

A. Agreement as a Reliability Signal

The central finding is the asymmetry in Table IV: 91% of test samples are unanimous and almost all of them are correct (0.50% error), while the remaining 9% contain 39 of the 41 samples on which any model erred. Disagreement therefore identifies a small subset of inputs that deserves a second look, at no cost beyond running the models already trained. A simple policy follows from Tables VI and VII: act directly on unanimous recommendations, flag 3–1 splits as less certain, and refer no-majority cases to an expert or request additional soil testing.

B. Majority Voting Does Not Beat the Best Single Model

Voting with 98.61% accuracy on 431 samples is not better than the Decision Tree alone, which reaches 98.64% on all 440 samples. The comparison is further tilted against voting if both are restricted to the same 431 samples: the Decision Tree makes 4 errors there (about 99.07% accuracy) against 6 for the vote. In at least two of the 3–1 splits the Decision Tree was the lone correct model and was outvoted by three SVMs. The value of the ensemble is therefore the abstention signal (Fig. 2), not higher accuracy; the Decision Tree at full coverage (1.36% risk) is comparable to the three-model majority at 97.95% coverage (1.39% risk), while unanimity attains lower risk (0.50%) at lower coverage.

C. Error Patterns

Rice and jute are the most frequently confused pair for the linear SVM, and the untuned polynomial SVM collapses many predictions to maize. We did not examine the feature distributions behind these patterns, so we regard overlapping feature ranges as a hypothesis for future work rather than a finding.

D. Limitations

(i) Results come from one random, non-stratified split of 440 test samples; a single sample changes accuracy by 0.23 points, and we did not test whether differences among the Decision Tree, linear SVM and RBF SVM are statistically significant. (ii) Hyperparameters were left at defaults, so the weakness of the polynomial SVM should not be read as an intrinsic property of that kernel. (iii) The benchmark is clean, balanced and widely used, and near-ceiling accuracy is common on it; we did not verify how its labels were collected, and the numbers may not transfer to field data. (iv) With four models, the agreement count has only three levels, so the coverage–risk curve has two operating points. (v) Models that agree on a wrong label cannot be detected (2 of 401 unanimous samples). (vi) No external dataset was used.

VI. Conclusion and Future Work

We examined whether disagreement among four classifiers marks less reliable crop recommendations, and found that it does: disagreement cases had markedly higher error than agreement cases, both by the any-model-wrong criterion and for every individual model. On a 440-sample hold-out set, unanimous predictions covered 91.14% of inputs with 0.50% error, and disagreement captured 39 of the 41 samples on which any model erred. Majority voting with a three-of-four threshold reached 97.95% coverage and 1.39% risk, but did not outperform the Decision Tree used alone, so its practical value lies in abstention and review flagging.

Future work should (a) use repeated stratified cross-validation or bootstrap confidence intervals, (b) tune hyperparameters with nested cross-validation, (c) enlarge the pool with tree ensembles and boosting to obtain a finer coverage–risk curve, (d) compare agreement-based abstention with probability-based abstention, and (e) validate on independent field data.

VII. References

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